2.24 Sound_Sensor_Example UNO

Sound_Sensor_Example UNO

Overview

In this lesson, you will learn how to use a sound sensor module. This module has two outputs:

- AO: analog output, real-time output voltage signal of microphone
- DO: when the intensity of the sound reaches a certain threshold, the output is a high or low level signal. The threshold sensitivity can be achieved by adjusting the potentiometer.

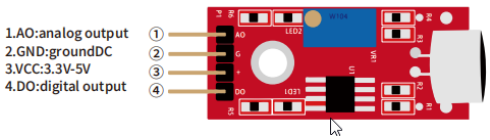
To make sure the microphone can detect your voice normally, please try to change its sensitivity by turning the blue precise potentiometer on the module. Given to its preciseness, you need to turn 10 times 360 degrees the potentiometer to start seeing some change.

Component Required:

(1) x Elegoo Uno R3

(1) x Sound sensor module

(4) x F-M wires (Female to Male DuPont wires)



Component Introduction

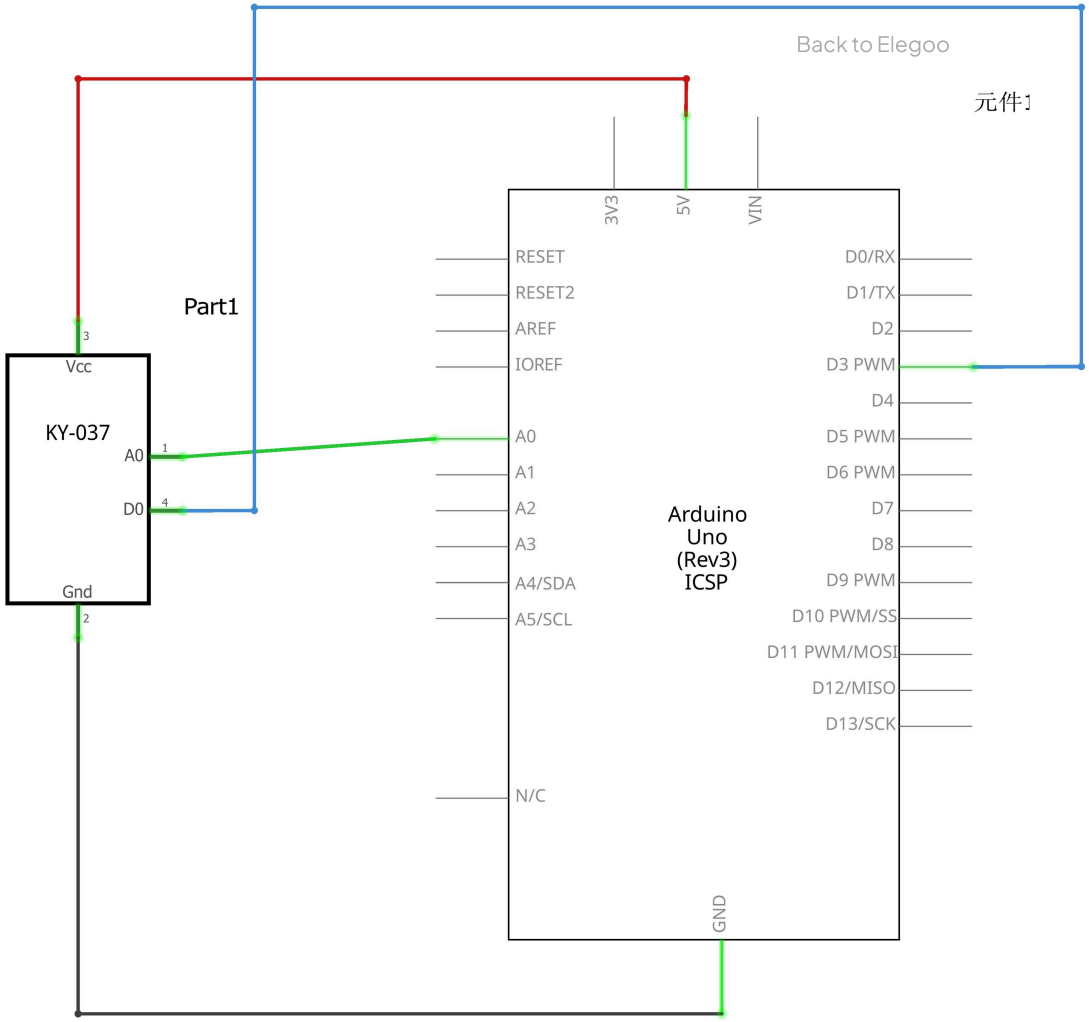
The sound sensor module provides an easy way to detect sound and is generally used for detecting sound intensity. This module can be used for security, switch, and monitoring applications. Its accuracy can be easily adjusted for the convenience of usage. It uses a microphone which supplies the input to an amplifier, peak detector and buffer. When the sensor detects a sound, it processes an output signal voltage which is sent to a microcontroller then performs necessary processing

These microphones are widely used in electronic circuits to detect minor sounds or air vibrations which in turn are converted to electrical signals for further use. The two legs as shown in the image above are used to make electrical connection with the circuit.

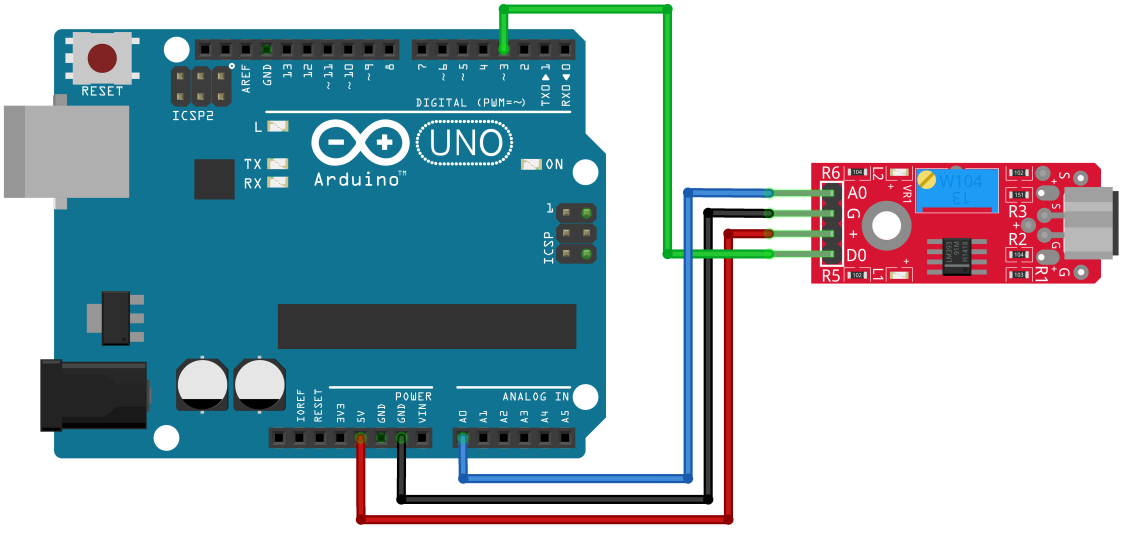


A solid conducting metal body encapsulates the various parts of the microphone. The top face is covered with a porous material with the help of glue. It acts as a filter for the dust particles. The sound signals/air vibrations pass through the porous material and falls on the diaphragm through the hole shown in the image above.

Connection Schematic



Wiring diagram



Code

You can click the blue text link to download the program file to your local device, and double-click the file to open it after the download is complete. Please note: Before opening the file, ensure that you have installed the Arduino IDE development environment and completed the installation of relevant components such as the board support package and driver corresponding to the UNO R3. If you have any questions about this operation process, you can refer to the "part 1" chapter of the document for detailed guidance.

DOWNLOAD sound_sensor.ino

This code demonstrates how to use a sound sensor with Arduino. It reads both analog and digital values from the sensor, displays the analog values on the serial monitor, and uses the built-in LED to indicate when sound is detected.

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Click the Serial Monitor button to turn on the serial monitor. The basics about the serial monitor are introduced in details in tutorial 4 in part 2.

```
int sensorAnalogPin = A0; // Select the Arduino input pin to accept the Sound Sensor's analog output
int sensorDigitalPin = 3; // Select the Arduino input pin to accept the Sound Sensor's digital output
int analogValue = 0; // Define variable to store the analog value coming from the Sound Sensor
int digitalValue; // Define variable to store the digital value coming from the Sound Sensor
int Led13 = 13; // Define LED port; this is the LED built in to the Arduino (labled L)
```

Variables and Definitions:

- `int sensorAnalogPin = A0;` : Arduino analog pin A0 connected to the sound sensor's analog output
- `int sensorDigitalPin = 3;` : Arduino digital pin 3 connected to the sound sensor's digital output
- `int analogValue = 0;` : Variable to store the analog value from the sound sensor
- `int digitalValue;` : Variable to store the digital value from the sound sensor
- `int Led13 = 13;` : Arduino built-in LED (labeled L)

```
void setup()
{
Serial.begin(9600); // The IDE settings for Serial Monitor/Plotter (preferred) must match this speed
pinMode(sensorDigitalPin,INPUT); // Define pin 3 as an input port, to accept digital input
pinMode(Led13,OUTPUT); // Define LED13 as an output port, to indicate digital trigger reached
}
```

Setup Function:

- `Serial.begin(9600);` : Initializes serial communication at 9600 baud rate
- `pinMode(sensorDigitalPin,INPUT);` : Sets the digital pin as input
- `pinMode(Led13,OUTPUT);` : Sets the built-in LED pin as output

```
void loop(){
analogValue = analogRead(sensorAnalogPin); // Read the value of the analog interface A0 assigned to digitalValue
digitalValue=digitalRead(sensorDigitalPin); // Read the value of the digital interface 3 assigned to digitalValue
Serial.println(analogValue); // Send the analog value to the serial transmit interface

if(digitalValue==HIGH) // When the Sound Sensor sends signla, via voltage present, light LED13 (L)
{
digitalWrite(Led13,HIGH);
}
else
{
digitalWrite(Led13,LOW);
}

delay(50); // Slight pause so that we don't overwhelm the serial interface
}
```

Loop Function:

- `analogValue = analogRead(sensorAnalogPin);` : Reads the analog value from the sound sensor
- `digitalValue=digitalRead(sensorDigitalPin);` : Reads the digital value from the sound sensor
- `Serial.println(analogValue);` : Prints the analog value to the serial monitor
- `if(digitalValue==HIGH)` : Checks if sound is detected
  - `digitalWrite(Led13,HIGH);` : Turns on the built-in LED if sound is detected
  - `digitalWrite(Led13,LOW);` : Turns off the built-in LED if no sound is detected
- `delay(50);` : Waits 50 milliseconds to avoid overwhelming the serial interface

Troubleshooting

- **No Data on Serial Monitor:** Check serial monitor baud rate is set to 9600
- **LED Not Responding:** Verify sensor connections to pins A0 and 3

- **Sensitivity Issues：** Adjust the potentiometer on the sound sensor module
- **No Sound Detection：** Ensure the sound sensor is properly powered and positioned
- **Serial Monitor Overflow：** Increase the delay time if the serial monitor is overwhelmed

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LCD 1602 →

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